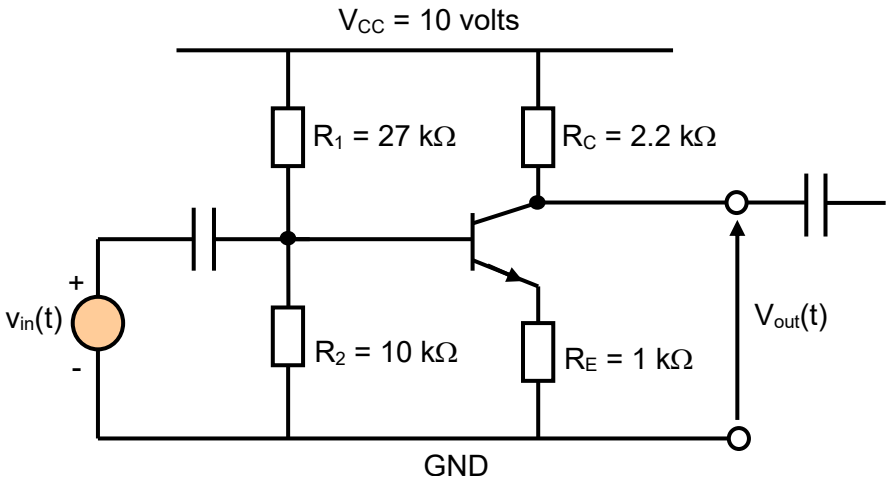
# BJT Amplifiers

**S. Le Goff**

Throughout this tutorial, we use silicon BJTs with $\beta_F = 100$, $V_{BE,on} \approx 0.7$ volt, and $V_{CE,sat} \approx 0.2$ volt.

**Question 1: Common-Emitter Amplifier with Negative Feedback**

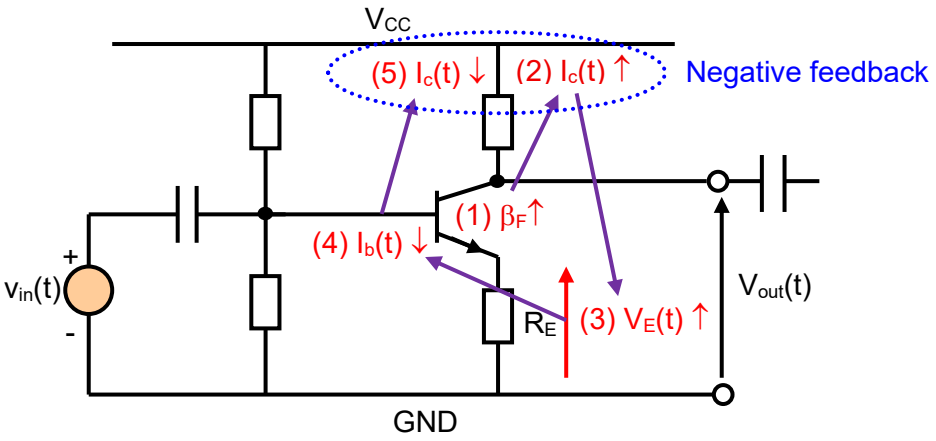
Consider the amplifier circuit depicted below.




The main difference with the amplifier circuit described in the lecture notes is that we have added an emitter resistor $R_E$. The presence of this resistor makes various parameters of the circuit, such as the DC operating points and the voltage gain, more stable, i.e. less dependent on the transistor characteristics.

To understand the role played by $R_E$, let us consider a simple example. Assume that the collector current $I_c(t)$ increases for any reason (e.g., because of an increase in $\beta_F$). In this case, the voltage across $R_E$, which is proportional to $I_c(t)$ since $I_c(t) \approx I_e(t)$, also increases. This results in a decrease in the base current $I_b(t)$ that tends to cancel the initial increase in $I_c(t)$ since $I_c(t) = \beta_F I_b(t)$.

This process, called *negative feedback*, does provide stability to the operation of the circuit.




Perform the DC and AC analysis of this amplifier.

We recall that the DC analysis of an amplifier consists of calculating the values of the DC currents $I_{B0}$ and $I_{C0}$ as well as the voltages $V_{BE0}$ and $V_{CE0}$. The main goal of the DC analysis is actually to determine whether or not the BJT(s) is/are properly biased. This analysis also allows us to calculate the value of the resistance $h_{11}$ using the equation

$$h_{11} \approx \frac{V_T}{I_{B0}}.$$

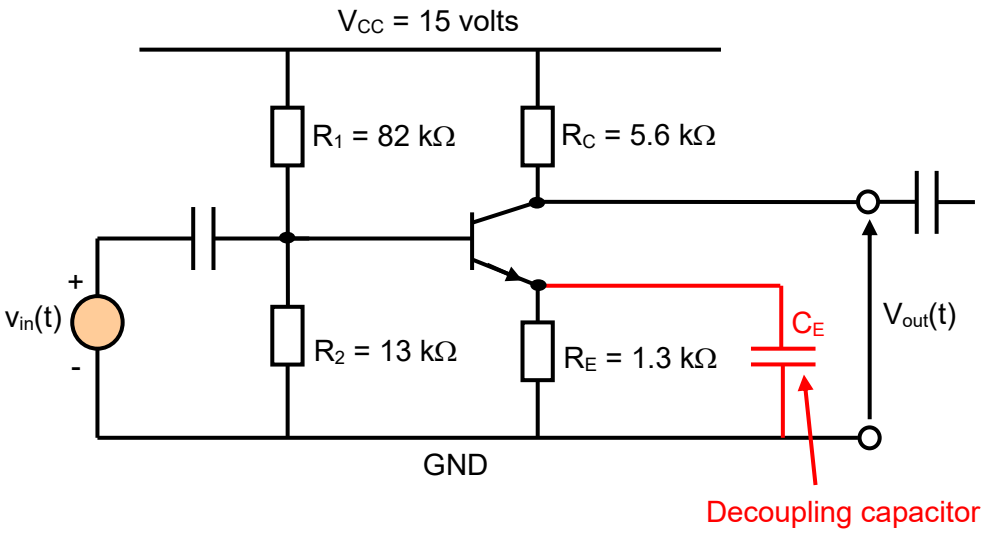
The AC analysis of an amplifier consists of finding its small-signal model and using the latter to calculate the values of the OC voltage gain $A_v$, input resistance $R_{in}$, and output resistance $R_{out}$.

Unless explicitly specified, we will not have to calculate the SC current gain $A_i$, the SC trans-conductance gain $A_g$, and the OC trans-resistance gain $A_r$, because we are here mainly interested in the operation of our circuits as voltage amplifiers.

Compare your results with those you would obtain without negative feedback, i.e. by using the same circuit with no emitter resistance ($R_E = 0$). Note that, if $R_E = 0$, the amplifier considered here is a classical CE amplifier similar to the one studied in the lecture notes.

**Question 2: CE Amplifier with Negative Feedback and Decoupling Capacitor**

Consider the circuit depicted below.




In Question 1, we have seen that the use of negative feedback has several advantages. These include the stabilisation of circuit parameters (e.g., DC operating points) which become less affected by changes in the transistor's characteristics. However, this is achieved at the expense of a considerable reduction in voltage gain.

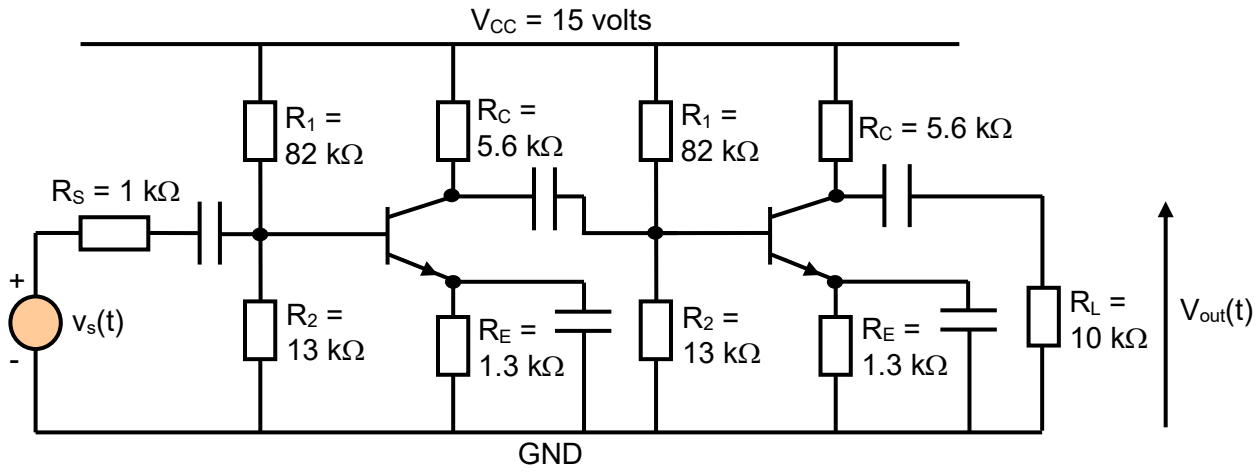
This fall in gain is a direct result of the negative feedback incorporated in the circuit. In some applications, this reduction in gain is unacceptable and a *decoupling capacitance* is used to reduce the amount of AC negative feedback while maintaining DC feedback. This increases the gain of the circuit but does not affect the DC feedback, which provides stability to the bias conditions of the circuit.

The decoupling capacitance $C_E$ is placed in parallel with the emitter resistance $R_E$, providing a low-impedance path for AC signals from the emitter to ground, but having no effect on the steady bias voltages.

Perform the DC and AC analysis of the amplifier shown above. For the AC analysis, assume that the decoupling capacitance $C_E$ is equivalent to a short circuit.

**Question 3: Cascaded Amplifier**

Consider the amplifier circuit depicted below.




This amplifier is obtained by *cascading* two CE amplifiers (using coupling capacitors) strictly identical to that studied in Question 2. The load resistance is $R_L$ = 10 kΩ, whereas the source resistance is $R_S$ = 1 kΩ.

Compute the real voltage gain $\tilde{A}_v = \frac{v_{out}(t)}{v_s(t)}$ of this *cascaded amplifier*.

What would be the value of $\tilde{A}_v$ if the load resistance $R_L$ was equal to 10 Ω instead of 10 kΩ?

**Question 4: Design of a CE Amplifier with Negative Feedback**

Design a CE amplifier with negative feedback. The specifications of this circuit are as follows:

- Power supply $V_{CC}$ = 10 volts;
- OC voltage gain $A_v \approx -9$;
- The maximum output voltage swing must be as large as possible;

- The input resistance $R_{in}$ must not be too small;
- The output resistance $R_{out}$ must not be too large.

**Question 5: CE Amplifier with Negative Feedback and Decoupling Capacitor**

Consider once again the amplifier circuit studied in Question 2 of this tutorial.

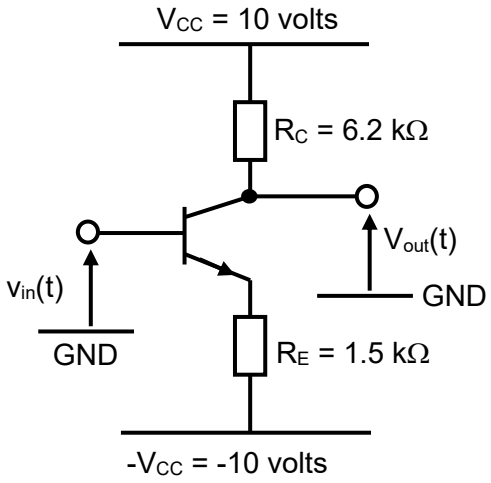
Find the expression of the OC voltage gain of this circuit by taking, this time, into account the decoupling capacitance $C_E$, rather than replacing it with a short circuit. To this end, assume that the input voltage $v_{in}(t)$ is a sinusoidal signal with frequency $\omega$.

What happens if $\omega << \frac{1}{R_E C_E}$?

What happens if $\omega >> \frac{1}{R_E C_E}$?

**Question 6: CE Amplifier with Symmetrical Power Supplies**

Consider the CE amplifier circuit depicted below. Unlike the circuits we have studied so far, this amplifier uses two symmetrical power supplies $V_{CC}$ = 10 volts and $-V_{CC}$ = - 10 volts.



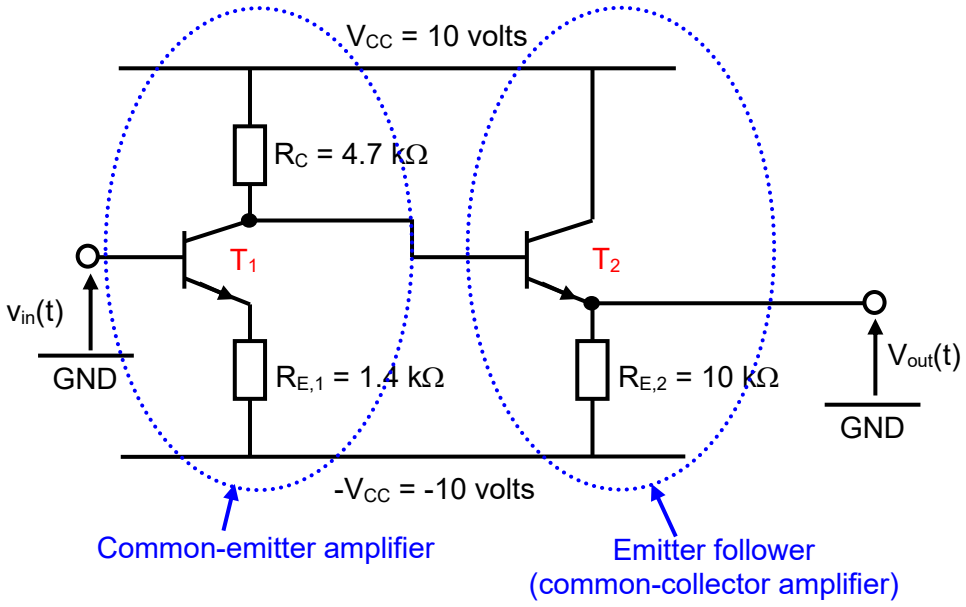
This arrangement provides several advantages such as a simplification of the biasing circuitry and a significant increase in the input resistance.

Perform the AC analysis of this amplifier. In particular, show that its input resistance is much larger than those of the amplifiers previously studied in Tutorial 3. For your computations, assume that $h_{11}$ = 2 kΩ.

**Question 7: CE Amplifier with an Output Stage**

Consider the amplifier circuit depicted below.

This circuit is designed by cascading a CE amplifier similar to that studied in Question 1 with an *emitter follower* (also called *common-collector amplifier*) acting as an output stage. This circuit is often referred to as a CE-CC configuration.




The collector of T2 is connected directly to the positive voltage supply. The input signal is applied between the base and ground, while the output signal is measured between the emitter and ground, which explains why such arrangement is called a common-collector (CC) amplifier.

Perform the AC analysis of this amplifier. In particular, show that its output resistance is much lower than that of a simple CE amplifier (thanks to the use of a CC amplifier at the output stage). For your computations, assume that $h_{11,T1} = h_{11,T2} = 2\ k\Omega$.

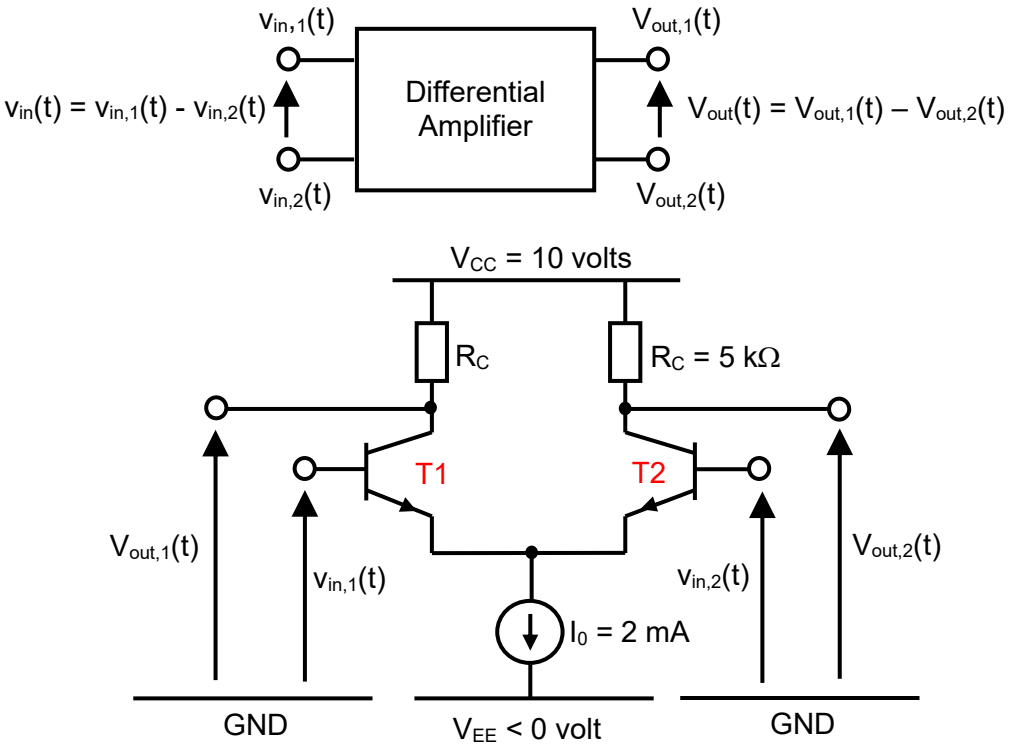
**Question 8: Differential Amplifier**

The amplifiers considered so far take as their input a single voltage that is measured with respect to some reference voltage, which is the “ground” or “earth” reference point of the circuit (0 volt).

Some amplifiers have not one but two inputs and produce an output proportional to the difference between the voltages on these inputs. Such amplifiers are called *differential amplifiers*.

Consider the differential amplifier circuit depicted below. The input voltages $v_{in,1}(t)$ and $v_{in,2}(t)$ are AC input voltages. They do not contain any DC component.

On the other hand, the output voltages $V_{out,1}(t)$ and $V_{out,2}(t)$ are output voltages that may contain a DC component, i.e. $V_{out,1}(t) = V_{OUT,1} + v_{out,1}(t)$ and $V_{out,2}(t) = V_{OUT,2} + v_{out,2}(t)$.



Note that the four voltages $v_{in,1}(t)$, $v_{in,2}(t)$, $V_{out,1}(t)$, and $V_{out,2}(t)$ are measured with respect to ground.

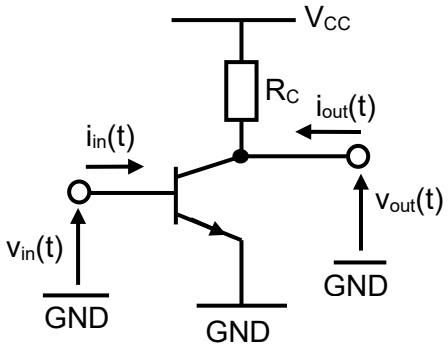
The ability to amplify the difference between two input signals is often extremely useful, for instance when some noise is added to both input voltages $v_{in,1}(t)$ and $v_{in,2}(t)$. If the noise that is added to $v_{in,1}(t)$ is very similar to that added to $v_{in,2}(t)$, which is often the case in practice, this noise will be essentially discarded since it will not be amplified by the differential amplifier.

This is not what happens with a traditional amplifier (single input voltage) for which any noise added to the input signal is unfortunately amplified along with the wanted signal.

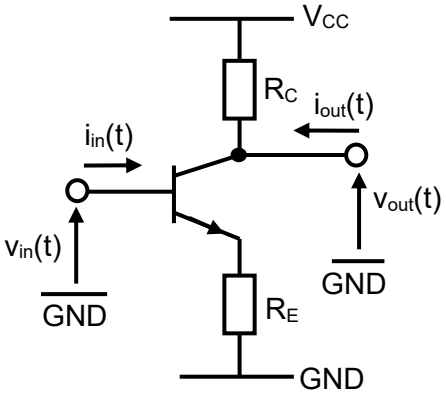
Perform the DC and AC analysis of the differential amplifier depicted below. Assume the use of two strictly identical silicon BJTs.

**Question 9: Characteristics of Generic BJT Amplifiers**

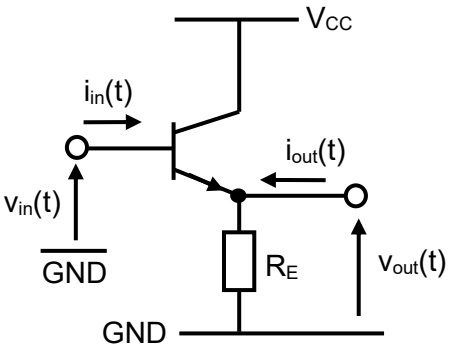
Consider the five generic amplifiers depicted below.



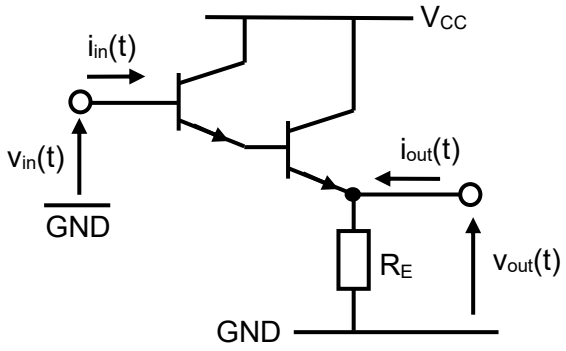

(a) Common-emitter amplifier



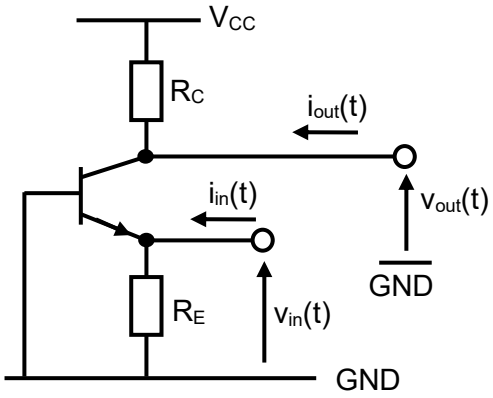

(b) Common-emitter amplifier with negative feedback




(c) Common-collector amplifier




(d) Darlington-pair amplifier




(e) Common-base amplifier

The term “generic” refers to the fact that no biasing circuitry is added to the circuit since we are only concerned here with the basic AC characteristics of the circuit.

In practice, one would of course have to take into account the biasing resistances in the computations (as we have done so far in all previous exercises).

Find the equivalent voltage amplifier model of each generic amplifier circuit.

You need to be careful as some of these amplifiers do not have a unilateral model. Therefore, to be on the safe side, it will be necessary to determine the expression of the reverse OC voltage gains for each amplifier.

In addition, find the equivalent current amplifier model of each circuit. To save time, note that it can be demonstrated that circuits that have a unilateral voltage amplifier model also have a unilateral current amplifier model with identical input and output resistances.

**- END -**